Setting Up the world Database

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This document describes `world` sample database installation, structure, usage, and history.

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Document generated on: 2026-08-04 (revision: 84779)

1 Preface and Legal Notices

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2 Installation

To enable MySQL users to perform manipulation of the `world` sample database using MySQL, the data set is available as a set of three tables:

- `country`: Information about countries of the world.
- `city`: Information about some of the cities in those countries.
- `countrylanguage`: Languages spoken in each country.

The `world.sql` file containing the `world` database is available for download at https://dev.mysql.com/doc/index-other.html. The file is provided as a compressed `tar` file or Zip archive.

To install the `world` sample database, follow these steps:

1. Download the installation archive to a temporary location such as `C:\temp\` or `/tmp/` and unpack it. Unpacking the archive results in a directory named `world-db` containing a single file named `world.sql`.
2. Connect to the MySQL server using the `mysql` command-line client with the following command:

   ```
   $> mysql -u root -p
   ```

   Enter your password when prompted. A non-`root` account can be used, provided that the account has privileges to create new databases.
3. Execute the `world.sql` script to create the database structure and insert the data using the following command:

   ```
   mysql> SOURCE C:/temp/world.sql;
   ```

   Replace the path to the `world.sql` file with the actual path on your system.

   > **Note**
   >
   > On Windows, use slashes rather than backslashes when executing the `SOURCE` command.

4. To confirm that the sample `world` database is installed correctly, execute the following statements. You should see output similar to that shown here.

   ```
   mysql> USE world;
   Database changed

   mysql> SHOW TABLES;
   +-----------------+
   | Tables_in_world |
   +-----------------+
   | city            |
   | country         |
   | countrylanguage |
   +-----------------+
   3 rows in set (0.00 sec)

   mysql> SELECT COUNT(*) FROM city;
   +----------+
   | COUNT(*) |
   +----------+
   | 4079     |
   +----------+
   1 row in set (0.02 sec)

   mysql> SELECT COUNT(*) FROM country;
   +----------+
   | COUNT(*) |
   +----------+
   | 239      |
   +----------+
   1 row in set (0.00 sec)
   ```

You now have the `world` sample database installed.

Another popular sample database is the Sakila database. For additional details, see http://dev.mysql.com/doc/sakila/en/.

3 History

September 2016

Prior releases used mixed-case table names. Because MySQL Shell is case-sensitive, table names are changed to lowercase.

December 2019

The `world.sql` file was changed to address several issues:

- The `world` database used `latin1`, but MySQL as of 8.0 uses a default character set of `utf8mb4`. Converting the database to use `utf8mb4` brings it up to date with MySQL 8.0, while retaining compatibility with older series.

  Conversion of the database to `utf8mb4` was done as follows:

  ```
  USE world;

  -- turn off foreign key checking; otherwise, the ALTER TABLE
  -- statements fail with incompatible foreign key errors.
  SET SESSION foreign_key_checks=0;

  -- convert database and tables to utf8mb4
  ALTER DATABASE world CHARACTER SET utf8mb4;
  ALTER TABLE city CONVERT TO CHARACTER SET utf8mb4;
  ALTER TABLE country CONVERT TO CHARACTER SET utf8mb4;
  ALTER TABLE countrylanguage CONVERT TO CHARACTER SET utf8mb4;

  -- re-enable foreign key checking
  SET SESSION foreign_key_checks=0;
  ```

- MySQL Shell requires a character set of `utf8mb4` for X Protocol connections. Using such a connection to load a `latin1`-encoded `world.sql` file produces errors.

  Using `mysqldump` with options of `--default-character-set=utf8mb4` to set the character set and `--set-charset` so the dump includes `SET NAMES` for that character set writes a `utf8mb4`-encoded dump file. Changing the encoding of `world.sql` to `utf8mb4` permits it to be loaded without errors in MySQL Shell using X Protocol connections.

- MySQL 8.0.17 deprecates these features:

  - Number of digits in floating-point column definitions; for example, `FLOAT(10,2)`.
  - Display width in integer column definitions; for example, `INT(10)`.

  The `world.sql` file used both features, causing load warnings in MySQL 8.0.17 and higher.

  To avoid number of digits in `FLOAT` columns, it is necessary to manually alter the relevant columns. That was done as follows, by altering `FLOAT(M,D)` to `DECIMAL(M,D)`:

  ```
  USE world;

  ALTER TABLE country
    MODIFY SurfaceArea DECIMAL(10,2) NOT NULL DEFAULT '0.00',
    MODIFY LifeExpectancy DECIMAL(3,1) DEFAULT NULL,
    MODIFY GNP DECIMAL(10,2) DEFAULT NULL,
    MODIFY GNPOld DECIMAL(10,2) DEFAULT NULL;
  ALTER TABLE countrylanguage
    MODIFY Percentage DECIMAL(4,1) NOT NULL DEFAULT '0.0';
  ```

  To produce integer column definitions without display widths in the dump file, it suffices to use `mysqldump` from MySQL 8.0.19 or higher.